An Examination of International Cuisines through Unsupervised Learning

Source: https://medium.com/data-science/an-examination-of-international-cuisines-through-unsupervised-learning-93c8b56d1ea0

How is machine learning used in this environment ...

In this study, a data analyst – Ben Sturm – sought to analyse data on recipes to find relationships between different cuisines.

Ben obtained data from 12,000 different recipes which could be attributed to 25 different national cuisines.

The first step was to format the data in such a way as to be able to use it in algorithms. This involved quite a lot of work, and involved **Natural Language Processing (NLP)**

The data came as JSON, but needed to be converted to a pandas dictionary for use in a data frame.

There was a lot of cleaning to do to get consistency of data. For instance, for each recipe, he needed a complete list of ingredients, but in the following ways:

- Stripping out the most common ingredients like salt or water
- Ensuring each ingredient was consistently listed, by making consistent hyphenations or plurals

With the data cleaned and prepared, Ben could then start using unsupervised learning algorithms.

He first used **k-Means Clustering** but this did not prove to be very useful – he was unable to identify why each cluster was being formed.

He then moved on to **principal component analysis (PCA)** – looking at the two principal ingredients of each recipe.

Again, this was initially messy.

But when he refined it to be grouped by cuisine, and plotting the centroid value for each, then clear clusters were evident.

And these were geographic e.g. Asian food shared similar principal ingredients, as did North European cuisines.

Not only that, but he was able to see which ingredients were *least* associated with each other ingredient. As an example if a dish contains onion, it's quite likely to contain garlic, but very unlikely to contain milk.

He then moved onto **Latent Dirichlet Allocation (LDA)**

This enabled him to do topic modelling. What he hypothesised was that he could group ingredients by cuisines

What actually was seen is that ingredients are more associated with a type of dish than a cuisine. E.g. sugar is found more often in desserts than in any one type of cuisine.

…and what does it do that humans can't (or would be too time intensive)?

It's clear that, with 12,000 recipes, even after the cleaning of the data, this would be a daunting task for a human.

A human would have to work through every possible permutation, and that is not even feasible.

A human too would come with their own ideas and associations.

In the case of **PCA** results, it way well be that a human intuition would help them get to the idea that Asian foods are more closely related than to other cuisines

But this may have hindered them when it came to the **LDA** results – it needs to work of the machine to dispel initial human assumptions, and show a quite different category of results.

The machine learning showed its value in the **k-means Clustering** too. Although this didn't generate a useful result, it did so quickly. A human would have had to invest a huge amount of – ultimately wasted – time to reach the same conclusion.